BAIO LEARNING SYSTEM • POWER PACK

PRINCIPLES OF INHERITANCE AND VARIATION

PU-II / Grade 12 Biology • Comprehensive & Analytical Study Material

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Board & NEET Analytics | Mendelian & Non-Mendelian Genetics | Pedigree & Chromosomal Syndromes

1. Fundamentals of Genetics & Technical Terminology

Genetics is the branch of biological science dealing with the principles and mechanisms of **inheritance** (transmission of traits from parents to progeny) and **variation** (the degree of phenotypic and genotypic difference between progeny and parents).

Core Genetic Terminology

- **Gene (Factor):** Functional unit of inheritance containing specific nucleotide sequences encoding a particular trait.
- **Alleles:** Alternative forms of the same gene situated at identical loci on homologous chromosomes.
- **Homozygous vs. Heterozygous:** Organism possessing identical alleles (*TT, tt*) vs. dissimilar alleles (*Tt*) for a specific gene.
- **Phenotype vs. Genotype:** Physical observable manifestation vs. underlying genetic complement of an organism.
- **Back Cross:** Cross between an F_1 hybrid and either of its parents.
- **Test Cross:** Cross of an individual of unknown dominant phenotype with a **homozygous recessive parent** (*_?* × *tt*). Used analytically to determine zygosity (homozygous dominant yields 100% dominant; heterozygous yields a **1:1 ratio**).
- **Out Cross:** Cross between an F_1 hybrid and its homozygous dominant parent.

2. Mendel's Experiments & The Seven Pairs of Contrasting Traits

Gregor Johann Mendel conducted hybridization experiments on the garden pea (*Pisum sativum*) for seven years (**1856–1863**), applying mathematical logic and large sample statistical analysis.

Character	Dominant Trait	Recessive Trait
Stem Height	Tall	Dwarf
Flower Colour	Violet / Purple	White
Flower Position	Axial	Terminal
Pod Shape	Inflated / Full	Constricted
Pod Colour	Green	Yellow
Seed Shape	Round	Wrinkled
Seed (Cotyledon) Colour	Yellow	Green

3. Monohybrid Cross & Mendel's Fundamental Laws

Cross involving a single pair of contrasting traits (e.g., Plant Height: $TT \times tt$):

- **F_1 Progeny:** 100% Tall (Tt).
- **F_2 Generation (Selfing of F_1):** Phenotypic Ratio = **3 Tall : 1 Dwarf (3:1)** ; Genotypic Ratio = **1 TT : 2 Tt : 1 tt (1:2:1)** .

Mendelian Principles Derived from Monohybrid Cross

1. **Law of Dominance:** Characters are controlled by discrete particulate units called *factors* occurring in pairs. In a dissimilar pair, one factor dominates and masks the expression of the recessive factor.
2. **Law of Segregation (Law of Purity of Gametes):** Alleles do not blend. The two alleles of a gene pair separate during gamete formation (meiosis), such that each gamete receives only one factor. Homozygotes produce identical gametes; heterozygotes produce two types in equal proportions (50% each).

4. Deviations from Mendelian Dominance

Incomplete Dominance (Partial Dominance)

- Neither allele is completely dominant; F_1 phenotype is intermediate between parents.
- **Examples:** Flower colour in *Antirrhinum majus* (Snapdragon) and *Mirabilis jalapa* (4 O'clock plant) ($RR = Red$, $rr = White$, $Rr = Pink$); Andalusian fowl feather colour ($BB = Black$, $bb = White$, $Bb = Blue$).
- **F_2 Ratios:** Phenotypic & Genotypic ratios are **identical**: **1 : 2 : 1** .

Co-Dominance & Multiple Allelism

- **Co-dominance:** Both alleles express themselves fully and independently in heterozygotes (e.g., AB Blood Group, Roan coat colour in cattle RW, Sickle Cell trait $Hb^A Hb^S$).
- **Multiple Allelism:** Presence of >2 alleles for a gene locus in a population (e.g., ABO blood groups governed by gene I with alleles I^A, I^B, i). Produces **6 Genotypes** and **4 Phenotypes**.

Genotype	Antigen on RBC Surface	Antibodies in Plasma	Blood Group Phenotype
I^AI^A or I^Ai	Antigen A	Anti-B	**Group A**
I^BI^B or I^Bi	Antigen B	Anti-A	**Group B**
I^AI^B	Antigen A and Antigen B (Co-dominant)	Neither	**Group AB** (Universal Recipient)
ii	No Antigens	Anti-A and Anti-B	**Group O** (Universal Donor)

5. Pleiotropy vs. Polygenic Inheritance

Pleiotropy (Single Gene → Multiple Phenotypes)	Polygenic Inheritance (Multiple Genes → Single Phenotype)
• A single gene influences multiple unrelated phenotypic pathways. • **Examples:** 1. **Phenylketonuria (PKU):** Mutation in phenylalanine hydroxylase gene (chr 12) causes mental retardation, hypopigmentation of hair and skin. 2. **Starch synthesis in Pea seeds:** Gene B controls seed shape (Round vs. Wrinkled) and starch grain size (Large, Intermediate Bb, Small).	• Quantitative traits controlled by three or more additive gene pairs showing continuous gradient variation. • Environmental factors significantly influence expression. • **Examples:** 1. **Human Skin Colour:** Governed by 3 genes (A, B, C). $AABBCC$ = Darkest; $AaBbCc$ = Intermediate (Mulatto); $aabbcc$ = Lightest. 2. **Kernel colour in wheat** (Nilsson-Ehle).

6. Dihybrid Cross & Law of Independent Assortment

Cross between plants differing in two traits: Seed Shape (Round R vs Wrinkled r) and Seed Colour (Yellow Y vs Green y) → $RRYY \times rryy$.

- **F_1 Generation:** 100% Round Yellow ($RrYy$).
- **F_2 Phenotypic Ratio:** **9 Round Yellow : 3 Round Green : 3 Wrinkled Yellow : 1 Wrinkled Green (9:3:3:1)**.
- **F_2 Genotypic Ratio:** **1:2:1:2:4:2:1:2:1** (representing 9 distinct genotypes).
- **Law of Independent Assortment:** When two pairs of traits are combined in a hybrid, segregation of one pair of characters is independent of the other pair during gametogenesis (applies strictly to non-linked genes).

7. Chromosomal Theory of Inheritance & Linkage Dynamics

Chromosomal Theory of Inheritance (Sutton & Boveri, 1902)

Postulated that Mendelian factors (genes) reside on chromosomes. Chromosomes and genes both occur in pairs, segregate during gametogenesis (Meiosis I anaphase), and independently assort.

T. H. Morgan's Experiments on Drosophila melanogaster

- Morgan chose *Drosophila* ("Cinderella of Genetics") because: (1) easily cultured on synthetic medium, (2) brief 2-week lifecycle, (3) high progeny output, (4) clear sexual dimorphism, and (5) distinct morphological markers under low magnification.
- **Linkage:** Physical association of genes located on the same chromosome that tend to be inherited together, preventing independent assortment.
- **Recombination:** Generation of non-parental gene combinations via crossing over during pachytene of Meiosis I.
- **Morgan's Dihybrid Cross Findings:**
  - **Cross A (Body colour & Eye colour):** y and w are tightly linked → **98.7% Parental : 1.3% Recombinant**.
  - **Cross B (Eye colour & Wing size):** w and m are loosely linked → **62.8% Parental : 37.2% Recombinant**.
- **Genetic Mapping (Alfred Sturtevant):** Used recombination frequency as a direct measure of physical map distance: ***1% ext{ recombination} = 1 ext{ map unit (cM - centiMorgan)}***.

8. Mechanisms of Sex Determination

Type	Male Complement	Female Complement	Organism Examples
XX - XY Mechanism (Male Heterogamety)	$AA + XY$ (produces 50% X, 50% Y sperm)	$AA + XX$ (homogametic, 100% X eggs)	Humans, *Drosophila*
XX - XO Mechanism (Male Heterogamety)	$AA + XO$ (produces 50% X, 50% nil sperm)	$AA + XX$ (homogametic)	Grasshopper, Insects
ZW - ZZ Mechanism (Female Heterogamety)	$AA + ZZ$ (homogametic)	$AA + ZW$ (produces 50% Z, 50% W ova)	Birds, Fowls, Reptiles
Haplodiploidy	Haploid ($n = 16$, developed parthenogenetically)	Diploid ($2n = 32$, from fertilized egg)	Honey bees (Drones have no father/ sons, but have grandfather/grandsons)

9. Pedigree Analysis & Mendelian Genetic Disorders

Pedigree Analysis: Study of the inheritance of specific traits over successive ancestral generations using standard symbols.

1. Autosomal Dominant Disorders

- Expressed in both homozygotes and heterozygotes; affects both sexes equally; vertical inheritance without generation skipping.
- **Examples: Myotonic Dystrophy**, Huntington's Chorea, Polydactyly, Brachydactyly.

2. Autosomal Recessive Disorders

- Expressed only in homozygous state (aa); unaffected carrier parents (Aa); often unmasked by consanguineous marriages.
- **Sickle Cell Anaemia:** Point mutation (transversion) in β-globin gene on **Chromosome 11** ($GAG \rightarrow GUG$; codon 6 substitutes **Glutamic acid → Valine**). Polymerization of HbS under hypoxia distorts RBCs into rigid sickle shapes.
- **Thalassemia (Quantitative Hb Defect):**
  - **α-Thalassemia:** Deficient α-chain synthesis governed by ***HBA1*** and ***HBA2*** genes on **Chromosome 16**.
  - **β-Thalassemia:** Deficient β-chain synthesis governed by ***HBB*** gene on **Chromosome 11**.
- **Phenylketonuria (PKU):** Autosomal recessive inborn error of metabolism on **Chromosome 12**; accumulation of phenylpyruvic acid causing microcephaly, mental deficits, and hypopigmentation.

3. Sex-Linked Disorders (X-Linked Recessive)

- Show characteristic **Criss-Cross Inheritance** (transmitted from affected father to carrier daughter to grandson). High prevalence in males (X^dY); females require homozygous recessive state (X^dX^d).
- **Haemophilia (Royal Disease):** Deficiency of clotting Factor VIII (Haemophilia A) leading to prolonged clotting times and fatal internal haemorrhage.
- **Colour Blindness (Daltonism):** Mutation in cone pigment genes on X chromosome (8% in males vs 0.4% in females) causing inability to distinguish red and green colours.

10. Chromosomal Syndromes & Aneuploidies

Disorder	Karyotype & Cause	Salient Clinical Features
Down's Syndrome (Trisomy 21)	***47, XX / XY (+21)*** Nondisjunction of 21st autosome	Short stature, round flat face, prominent epicanthic fold, permanently open mouth with furrowed protruding tongue, single transverse palmar crease, congenital heart defects, psychomotor and mental retardation.
Klinefelter's Syndrome (Allosomal Trisomy)	***47, XXY*** Extra X chromosome in males	Overall masculine build with feminine physical features, tall stature, eunuchoid proportions, **Gynaecomastia** (breast development), azoospermia/hypogonadism, sterility.
Turner's Syndrome (Allosomal Monosomy)	***45, X0*** Absence of one X chromosome	Phenotypic female with short stature, webbed neck, broad shield-shaped chest with widely spaced nipples, **rudimentary streak ovaries**, amenorrhea, lack of secondary sexual characteristics, sterility.
Cri-du-Chat Syndrome	***46, 5p-*** Partial deletion of short arm of Chr 5	High-pitched cat-like infant cry (laryngeal hypoplasia), microcephaly, hypertelorism, severe mental retardation, and developmental delays.

11. High-Yield Exam Review & Analytical Ratios

Genetic Cross / Concept	Phenotypic Ratio	Genotypic Ratio
Monohybrid Cross (Complete Dominance)	3 : 1	1 : 2 : 1
Incomplete Dominance & Co-Dominance	1 : 2 : 1	1 : 2 : 1
Monohybrid Test Cross	1 : 1	1 : 1
Dihybrid Cross (Independent Assortment)	9 : 3 : 3 : 1	1:2:1:2:4:2:1:2:1
Dihybrid Test Cross	1 : 1 : 1 : 1	1 : 1 : 1 : 1